

Resource-Aware Keyword Spotting with MATLAB and MFCC Features

A MATLAB study of MFCC front ends, speaker-disjoint evaluation, and compact keyword-spotting models for resource-constrained audio systems.

Student	Lucas(Wenqi) Wang
Student ID	550552222
GitHub username	wenqiwang1314-dotcom
Repository	https://github.com/wenqiwang1314-dotcom/elec5305-project-550552222
Project site	https://wenqiwang1314-dotcom.github.io/elec5305-project-550552222/

Project Overview

This project investigates a compact keyword-spotting system that recognises ten spoken commands - down, go, left, no, off, on, right, stop, up, and yes - plus unknown speech and silence. Keyword spotting is the small-vocabulary front end used by voice-controlled devices before a larger speech recogniser is activated. It must be accurate enough to avoid false activations, but computationally small enough to run continuously on resource-constrained hardware. The proposed solution is a reproducible MATLAB pipeline covering audio preparation, Mel-frequency cepstral coefficient (MFCC) extraction, baseline classification, and later comparison with a depthwise-separable convolutional neural network (DSCNN). The design is inspired by Texas Instruments' 12-class Google Speech Commands example for the MSPM0G5187, while the present project remains a PC-side simulation until separate embedded deployment has been implemented and tested.

Background and Motivation

Always-on keyword spotting must balance recognition, memory, computation, and energy. MFCCs compactly describe the short-term spectral envelope on a perceptually motivated Mel scale. Prior work establishes MFCCs, compact CNNs, depthwise-separable CNNs, compression, and robust front ends as complementary design choices. This project therefore connects ELEC5305 concepts - framing, windowing, Fourier analysis, filterbanks, cepstral features, and objective evaluation - to a practical edge-audio application.

Proposed Methodology

The experiments will use Speech Commands v0.02, a public dataset containing 105,829 one-second spoken-word recordings at 16 kHz. The supplied official validation and testing lists will be respected so recordings from the same speaker do not leak across partitions. Ten commands will be retained as known classes. Other word folders will form the unknown class, and one-second regions from the background-noise recordings will form the silence class.

MATLAB will first standardise each clip to one second at 16 kHz. The acoustic front end will follow the TI example: 30 ms frames, 20 ms frame step, 40 Mel filters, and 10 MFCC coefficients. This produces 49 time frames by 10 coefficients for each utterance. A transparent k-nearest-neighbour classifier will provide a

smoke-test baseline and verify the complete data path without requiring a deep-learning toolbox. The next model will be a compact DSCNN that operates directly on the 49 by 10 feature map. Controlled experiments will compare the baseline and DSCNN using identical data partitions. A fixed-manifest front-end ablation now also compares coefficient count, frame step, pre-emphasis, cepstral mean normalization (CMN), and a literature-motivated log-Mel comparator while recording feature-map size and 10 dB robustness.

Evaluation will report overall test accuracy, per-class precision and recall, macro recall, and a confusion matrix. Special attention will be given to the unknown and silence classes because they determine false activations in a realistic always-on system. Runtime, parameter count, and estimated memory will be recorded for resource awareness. Repeated training seeds will be used before drawing model-ranking conclusions. Hardware compilation or energy measurements will be reported only if they are later performed on the named target under a separate, documented protocol.

Expected Outcomes

Deliverables are runnable MATLAB code, an exact manifest, acoustic figures, machine-readable metrics, a verified literature set, and a GitHub Pages site. The final target is a multi-seed DSCNN comparison with explicit accuracy, unknown rejection, robustness, computation, and memory costs.

Timeline (Weeks 1-13)

Weeks	Work package
1-2	Define the 12-class task, audit the dataset, and review literature.
3-5	Implement and verify audio loading, official splits, MFCC extraction, and the baseline.
6-9	Implement the DSCNN, establish repeated-seed training, and tune only on validation data.
10-11	Evaluate test performance, runtime, memory, robustness, and error patterns.
12-13	Finalise the report, reproducibility package, GitHub documentation, and demonstration.

References

1. P. Warden, "Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition," arXiv:1804.03209, 2018.
2. Y. Zhang, N. Suda, L. Lai, and V. Chandra, "Hello Edge: Keyword Spotting on Microcontrollers," arXiv:1711.07128, 2017.
3. S. B. Davis and P. Mermelstein, "Comparison of Parametric Representations for Monosyllabic Word Recognition in Continuously Spoken Sentences," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357-366, 1980, doi:10.1109/TASSP.1980.1163420.
4. T. N. Sainath and C. Parada, "Convolutional Neural Networks for Small-footprint Keyword Spotting," Proceedings of Interspeech, pp. 1478-1482, 2015, doi:10.21437/Interspeech.2015-147.
5. G. Tucker, M. Wu, M. Sun, S. Panchapagesan, G. Fu, and S. Vitaladevuni, "Model Compression Applied to Small-Footprint Keyword Spotting," Proceedings of Interspeech, pp. 1878-1882, 2016, doi:10.21437/Interspeech.2016-1393.
6. Texas Instruments, "Google Speech Command Recognition," Tiny ML Tensorlab 1.4.0 User Guide, 2026.
7. Y. Wang et al., "Trainable Frontend for Robust and Far-Field Keyword Spotting," ICASSP, 2017, doi:10.1109/ICASSP.2017.7953242.
8. P. M. Sorensen, B. Epp, and T. May, "A Depthwise Separable Convolutional Neural Network for Keyword Spotting on an Embedded System," EURASIP JASMP, 2020, doi:10.1186/s13636-020-00176-2.
9. D. Peter, W. Roth, and F. Pernkopf, "Resource-efficient DNNs for Keyword Spotting using Neural Architecture Search and Quantization," arXiv:2012.10138, 2020.

10. A. Osman et al., "TinyML Platforms Benchmarking," arXiv:2112.01319, 2021.
11. A. Almaini, J. Folz, and G. Ashour, "TinyML for Acoustic Anomaly Detection in IoT Sensor Networks," arXiv:2603.26135, 2026.

Preliminary Evidence

The first smoke test used 720 training and 240 held-out test utterances with a manual 5-nearest-neighbour classifier. It produced the required 49 by 10 MFCC map, found zero speaker overlap, and reached 39.17% accuracy and macro recall, above the 8.33% chance level. Silence recall was 100%, but unknown-word recall was 0%. On the identical manifest and classifier, adding utterance CMN raised clean accuracy to 47.08% at the same 490-element input; pre-emphasis reached 41.67%. Denser 98-frame inputs added little despite doubling or tripling the feature size. At 10 dB, every clean-trained variant fell to 8.33-10.42%, exposing noise mismatch as a more important next problem than increasing MFCC size. These are single-subset PC-side diagnostics, not DSCNN or embedded results.

MFCC Front-End Optimization

A controlled ablation keeps the 720/240 manifest and manual 5-NN fixed while changing only the acoustic front end. Clean-trained models are also tested with deterministic 10 dB additive noise. Feature elements are a transparent input-size proxy, not a target-RAM measurement.

Front end	Shape	Elements	Clean	10 dB
TI reference	49 x 10	490	39.17%	8.33%
13 MFCC	49 x 13	637	40.42%	8.33%
10 ms hop	98 x 10	980	40.83%	8.33%
25/10 ms, 13 MFCC	98 x 13	1274	39.17%	8.33%
TI + pre-emphasis	49 x 10	490	41.67%	8.33%
TI + CMN	49 x 10	490	47.08%	10.00%
20-bin MFSC	49 x 20	980	28.75%	10.42%

CMN reaches 47.08% clean accuracy at the same 490-element input. The near-chance 10 dB results show that noise-aware training is a higher priority than increasing the feature map.

Appendix: Preliminary Real-Data Results

The following figures were generated by the submitted MATLAB smoke test from held-out Speech Commands v0.02 samples. The result is a 5-NN baseline, not a DSCNN or embedded-device measurement.

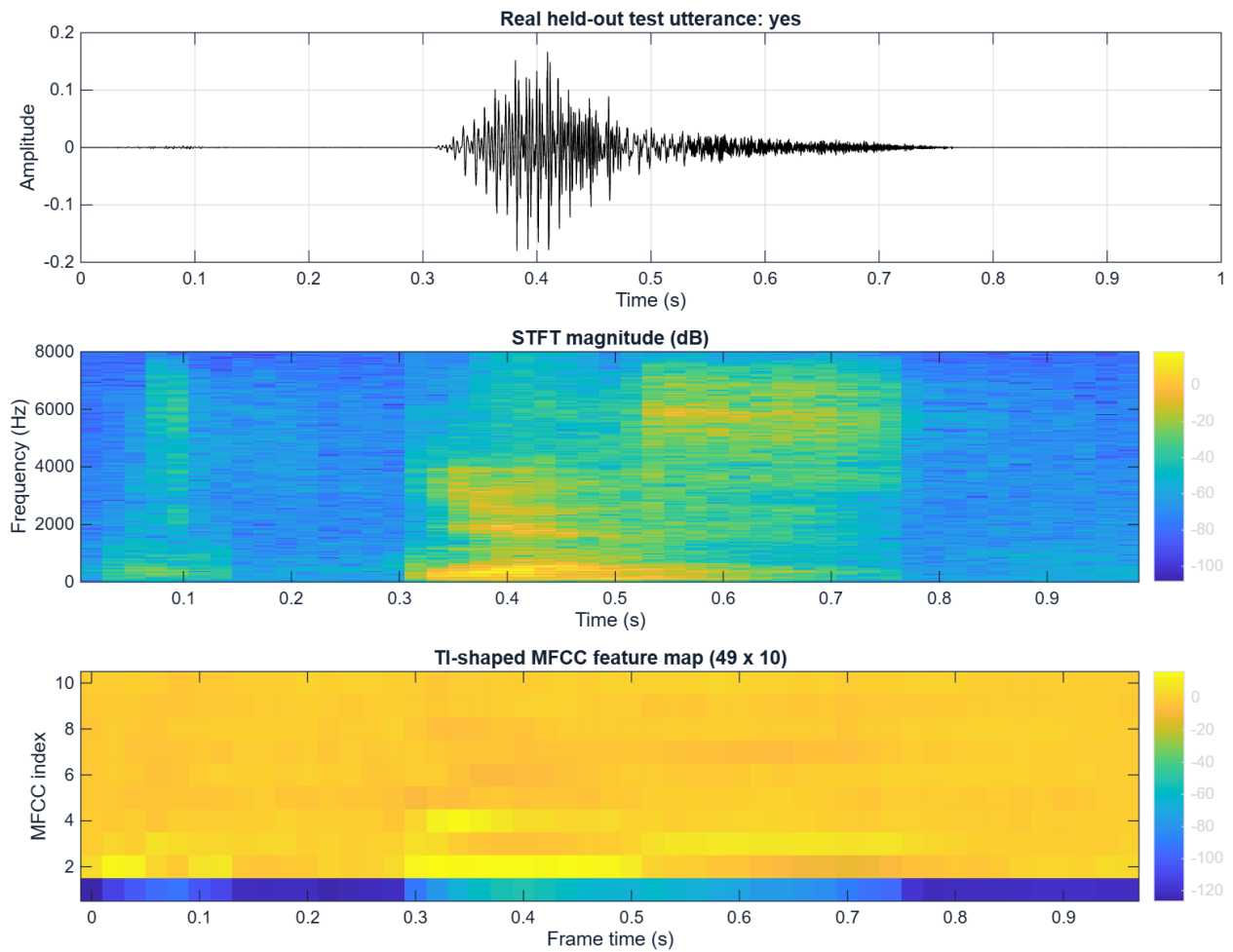


Figure A1. Real waveform, STFT, and TI-shaped 49 x 10 MFCC map.

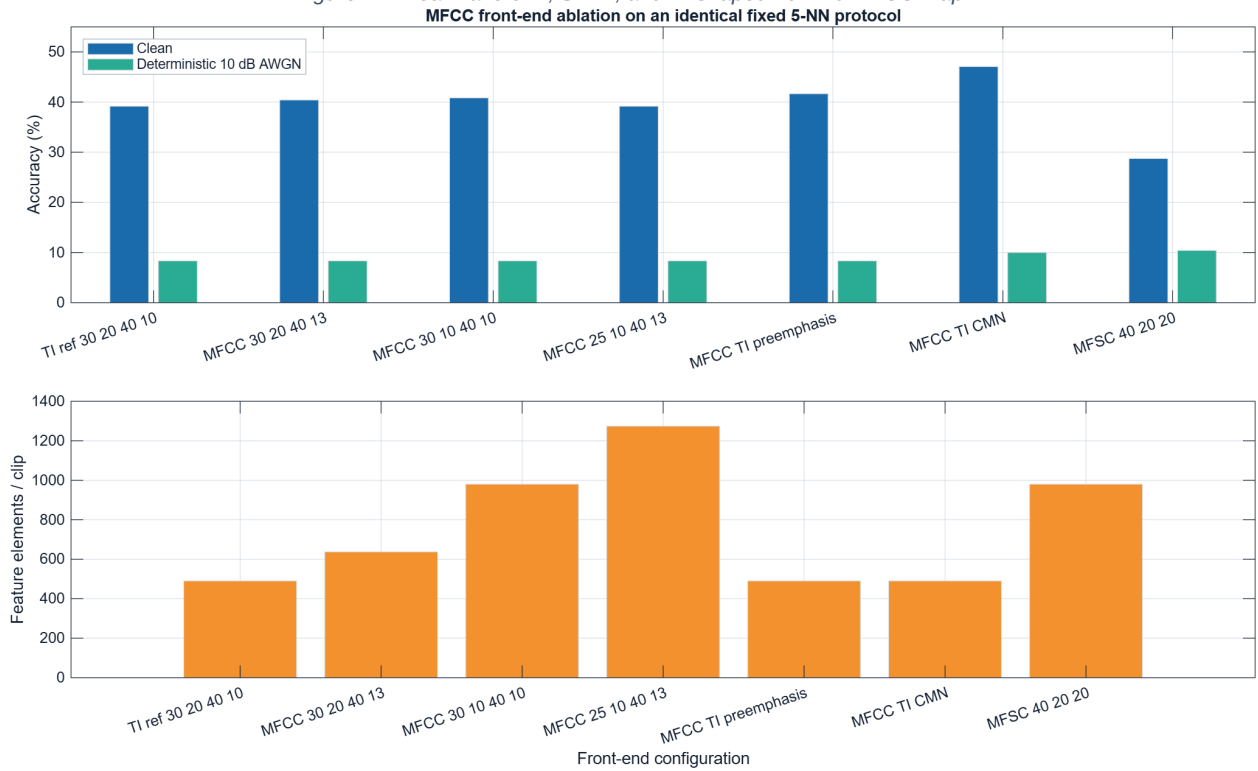


Figure A2. Controlled front-end accuracy and input-size comparison.

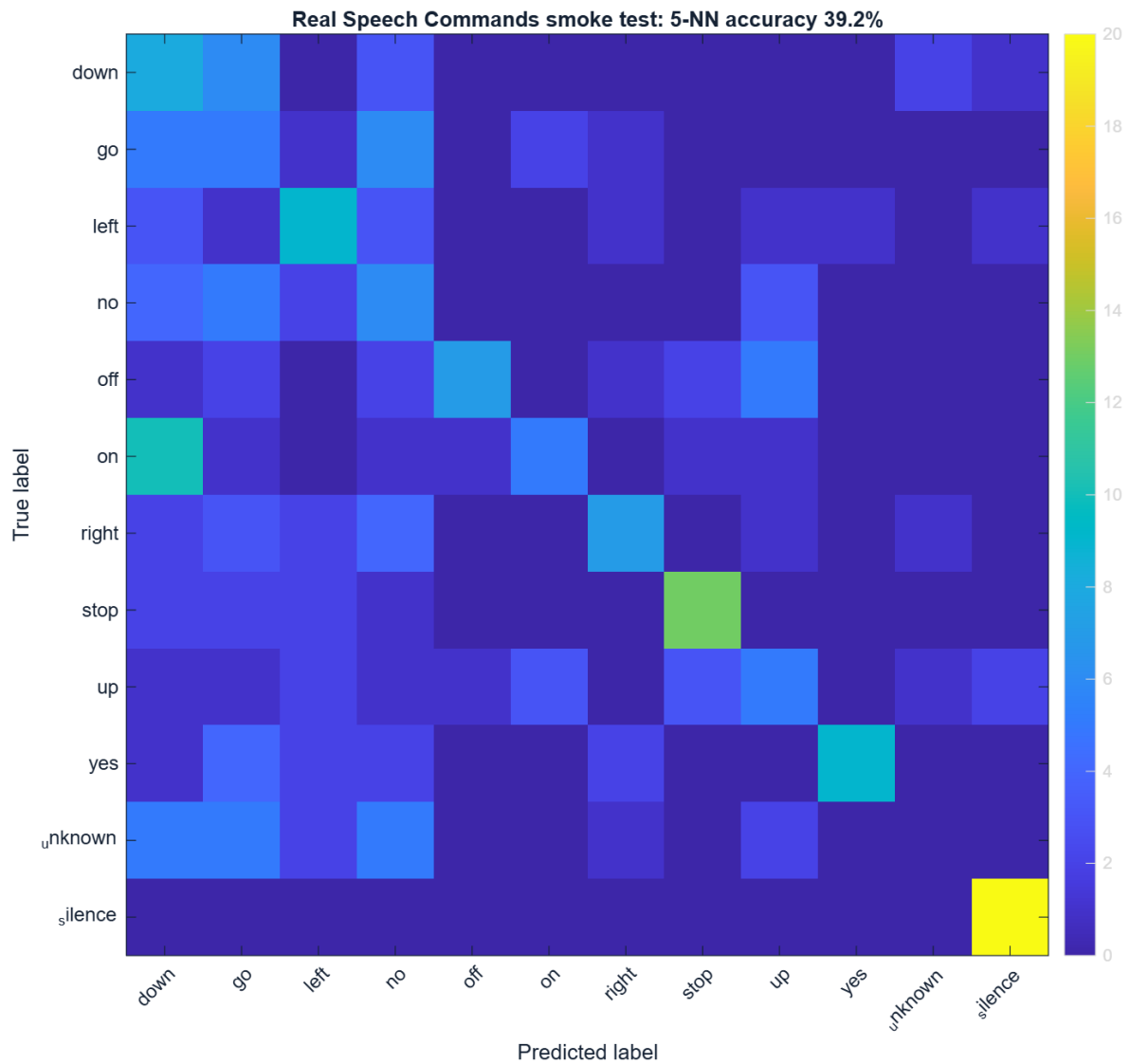


Figure A3. Speaker-disjoint 12-class smoke-test confusion matrix (240 test samples).